INCH-POUND

MIL-C-70461D (AR)
28 May 1991
SUPERSEDING
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4 April 1986

MILITARY SPECIFICATION

COMBUSTIBLE CARTRIDGE CASE ASSEMBLY PARTS FOR CHARGE, PROPELLING, 155MM, M203A1

This specification is approved for use by the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the component parts of the propelling charge designated as charge, propelling, 155MM, M203A1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and Supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-C-70462	- Charge Propelling, 155MM, M203A1 Loading, Assembly and Packing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-286	- Propellants, Solid, Sampling, Examination and Testing
MIL-STD-1168	- Lot Numbering of Ammunition

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbin Ave., Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9345091	- Case Cartridge Assembly 155MM, M203A1
9345095	- Rear, Cap
9345097	- Igniter Container
9345098	- Body, Molded
9345109	- Composition
9357034	- Composition
9390469	- Packing and Marking for Box, Fiberboard for Igniter Container and Rear Cap, for Case Cartridge Assembly, 155MM, M203A1
9390470	- Packing and Marking for Box, Fiberboard for Body, Molded or Wrapped Body Assembly for Case Cartridge Assembly, 155MM, M203A1
9390475	- Packing and Marking for Box, Wood, Cleated Plywood for Box, Fiberboard for Igniter Container and Rear Cap for Case Cartridge Assembly, 155MM, M203A1
9390476	- Packing and Marking for Box, Wood, Cleated Plywood for Box, Fiberboard for Body, Molded or Wrapped Body Assembly for Case Cartridge Assembly, 155MM, M203A1

(Copies of drawings, publications and other Government documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by contracting officer.)

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2.2 Non-Government publications. The following document(s) forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D638 - Test Method for Tensile Properties of Plastics

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. The constituent materials shall comply with the requirements of all applicable drawings and specifications.

3.2 Parts. The combustible cartridge case parts shall comply with all the requirements specified in applicable drawings and specifications.

3.3 Stability. The nitrocellulose in the combustible cartridge case parts shall not completely change the color of the methyl violet paper to salmon pink color in less than thirty-five (35) minutes when tested in accordance with 4.5.2.

3.4 Physical properties. The weights and tensile strength of combustible cartridge case parts shall meet the requirements of Table I.

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TABLE I
Weight and tensile strength requirements for combustible cartridge case parts.

Part Description	Drawing	Weight	Tensile Strength	Density (1)
Rear Cap	9345095	2.0 ± 0.25 oz	N/A	0.85 ± 0.12g/cc
Igniter Cont.	9345097	6.8 ± 0.49 oz	N/A	0.85 ± 0.12g/cc
Body, Molded	9345098	29.0 ± 1.64 oz	4600 ± 2000 psi	0.98 ± 0.12g/cc

Notes (1) Density is advisory

3.5 50 - lb static test. The assembly of igniter container and body shall pass the 50 - lb static test and no separation of igniter container from the assembly shall occur when tested in accordance with 4.5.1.

3.6 Assembly test. The mating surfaces between rear cap and igniter container shall be tight. The igniter container shall not be separated from the assembly when tested in accordance with 4.5.7.

3.7 Destructive pull test. The force of the adhesive bond between rear cap and igniter container shall be 200 lbs minimum when tested in accordance with 4.5.8.

3.8 Proving ground (see 4.5.9). Ballistic tests shall be performed for First Article (see 4.3.2) and the first lot of each new interfix at 70 degrees F, 145 degrees F and -65 degrees F. The cartridge case components, when assembled and fired as 155MM, M203A1 propelling charge shall be consumed such that there is no inert or burning combustible cartridge case residue in the gun tube.

3.9 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.9.1 First article test. Test samples shall be withdrawn from the first lot produced by the contractor and tested in accordance with all the requirements of this specification and with Table VII. Until the first article test has been performed and found to be acceptable, the contractor is in no way authorized by the Government to initiate second lot production unless otherwise directed by the contracting officer.

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3.10 Workmanship. All parts shall be fabricated in a thorough, workmanlike manner, and shall be free of molding flaws, cracks, warpage outside the material envelope, crevassing, soft spots, frayed edges, foreign material greater than 1/4 inch in any dimension.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as parts of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on the item:

a. First Article Inspection (4.3)
b. Quality Conformance Inspection (4.4)

Prior to dimensional examination of combustible cartridge case, components shall be conditioned for a minimum of twelve (12) hours at 23 degrees Centrigrade $\pm$ 2 degrees and at 50% $\pm$ 5% relative humidity.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample, as designated by the contracting officer for first article tests. The sample shall consist of parts which

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have been produced by the contractor or furnished by a supplier which have been manufactured using the same production process, procedures, molds, dies and equipment which will be used in fulfilling the contract. If parts are produced from more than one (1) mold or die, they shall be identified by a suitable mold or die mark. Samples shall be taken equally from each mold or die used in production. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to ensure that it conforms to the requirements of the contract, and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. (A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever a change occurs in manufacturing process, material used, or source of supply as to significantly affect product uniformity as determined by the Government). The first article sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawings	Quantity
Body, Molded	9345098	80
Rear Cap	9345095	80
Igniter, Container	9345097	80

In addition, whenever there is a lapse in production for a period in excess of 90 days, a first article sample, as directed by the contracting officer shall also be submitted as follows:

Part description	Drawings	Quantity
Body, Molded	9345098	50
Rear Cap	9345095	50
Igniter, container	9345097	50

4.3.2 Inspection to be performed. The first article parts consisting of quantity of 80 submitted in accordance with 4.3.1 will be subjected by the Government to any or all of the requirements of the applicable drawings and specifications and to the inspections of Table II. The first article parts consisting of quantity of 50 submitted in accordance with 4.3.1 will be subjected by the Government to any or all of the requirements of the applicable drawings and specifications and to the inspections of Table II with the exception of firing tests. All inspections listed in Table II shall be classified Major unless otherwise specified in referenced paragraphs and specifications.

4.3.3 Rejection. See MIL-A-48078.

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TABLE II. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Combustible Cartridge Case Components	SHEET 1 OF 1	DRAWING NUMBER See Below NEXT HIGHER ASSEMBLY	
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Body, Molded (Drawing 9345098) Examination for defects	50	3.2	4.4.2.1
	Igniter Container (Drawing 9345097) Examination for defects	50	3.2	4.4.2.2
	Rear Cap (Drawing 9345095) Examination for defects	50	3.2	4.4.2.3
	Combustible Cartridge Case Components (all) assembled as part of the M203A1 propelling charge, Firing Tests at 70°F, 145°F, and -65°F.	30	3.8	4.5.9

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078 and MIL-STD-1168. In addition, inspection lots of parts for case shall contain:

a. Nitrocellulose from one manufacturer and from not more than two lot numbers with the same interfix number.

b. Diphenylamine from one manufacturer and from not more than two lot numbers with the same interfix number.

c. Resin from one manufacturer and from not more than two lot numbers with the same interfix number.

d. Kraft from one manufacturer and not more than two lot numbers with the same interfix number.

e. Talc from one manufacturer and not more than two lot numbers with the same interfix number.

4.4.2 Examinations and tests. See MIL-A-48078.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawings and specification requirements utilizing as a minimum the conformance criteria specified. When cited in the classification of characteristics paragraphs, attributes sampling inspection shall be conducted in accordance with Table III below.

Table III
Attributes Sampling for Major and Minor Defects

Lot Size	Major Sample Size	Minor Sample Size
2 to 8	*	See note
9 to 15	*	5
16 to 25	*	5
26 to 50	32	5
51 to 90	32	12
91 to 150	32	12

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Table III
Attributes Sampling for Major and Minor Defects Continued...

Lot Size	Major Sample Size	Minor Sample Size
151 to 280	32	14
281 to 500	32	17
501 to 1200	74	20
1201 to 3200	74	24
3201 to 10000	87	30
10001 to 35000	109	36
35001 to 150000	124	40
150001 to 500000	156	40
500001 and over	187	40

* 100 Inspection

NOTE: Use 100% inspection when lot size is less than 5 units. Use sample size when lot size is greater than 5 units.

b. Alternative quality conformance provisions. Alternative quality conformance provisions, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc. may be used by the contractor when they provide as a minimum the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.9). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70461D (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Body, Molded		9345098
			NEXT HIGHER ASSEMBLY 9345103

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter	Table III	3.2	Gage
102	Inside diameter	Table III	3.2	Gage
103	Material thickness	Table III	3.2	Gage
104	Length	Table III	3.2	Gage
105	Straightness	Table III	3.2	Gage
106	Weight	Table III	3.4	4.5.6
107	Composition	4.4.3.5	3.2	4.5.3
108	Tensile strength	4.4.3.6	3.4	4.5.4
109	Density	4.4.3.7	3.4	4.5.5
110	Presence of ribs within 2.00 in.from rear end	Note 1	3.2	Gage
111	Diameter of holes (4 places)	Table III	3.2	Gage
112	Unequally spaced holes (4 places)	Note 1	3.2	Gage
113	Nitrocellulose stability	4.4.3.4	3.3	4.5.2
Minor				
201	Outside radius	Note 1	3.2	Gage
202	Improper chamfer (angle and length)	Table III	3.2	Visual
203	Evidence of poor workmanship	Table III	3.10	Visual

NOTES: 1/ Tool Controlled Dimensions - Whenever a new tool is introduced, 25 samples shall be checked for conformance with the dimension requirements for qualification of tool prior to production.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Igniter Container	1 OF 2	9345097
			NEXT HIGHER ASSEMBLY 9345103

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Maximum outer diameter of igniter container	Table III	3.2	Gage
102	Outer diameter of rear lip/rear cap interface	Table III	3.2	Gage
103	Length of container	Table III	3.2	Gage
104	Length of rear lip (0.60 in.)	Table III	3.2	Gage
105	Length of rear lip edge to rear body interface	Note 1	3.2	Gage
106	Length of maximum outer diameter	Note 1	3.2	Gage
107	Front outer diameter	Table III	3.2	Gage
108	Thickness of material	Table III	3.2	Gage
109	Weight	Table III	3.4	4.5.6
110	Composition	4.4.3.5	3.2	4.5.3
111	Density	4.4.3.7	3.4	4.5.5
112	Length from rear edge of body interface to center of boss (4 places)	Note 1	3.2	Gage
113	Diameter of bosses (4 places)	Note 1	3.2	Gage
114	Unequally spaced bosses (4 places)	Note 1	3.2	Gage
115	Length of flat surface of boss (4 places)	Note 1	3.2	Gage
116	Angle on boss (4 places)	Note 1	3.2	Gage
117	Nitrocellulose stability	4.4.3.4	3.3	4.5.2
Minor				
201	Improper chamfer (angle and length)	Table III	3.2	Visual
202	Radius of rear outer corner	Table III	3.2	Functional Volumetric Gage
203	Improper taper angle (7°)	Note 1	3.2	Gage

NOTES: 1/ Tool Controlled Dimensions - Whenever a new tool is introduced, 25 samples shall be checked for conformance with the dimension requirements for qualification of tool prior to regular production.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70461D (AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.2	Igniter Container		9345097
			NEXT HIGHER ASSEMBLY 9345103

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor	Continued...			
204	Improper taper angle (15°)	Note 1	3.2	Gage
205	Diameter of holes	Note 1	3.2	Gage
206	Number of holes	Note 1	3.2	Visual
207	Holes not properly spaced	Note 1	3.2	Gage
208	Diameter of rear flat portion	Table III	3.2	Functional Volumetric Gage
209	Evidence of poor workmanship	Table III	3.10	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Cap, rear	1 OF 1	9345095
			NEXT HIGHER ASSEMBLY 9345092

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter of cap	Table III	3.2	Gage
102	Inside diameter of cap	Table III	3.2	Gage
103	Material thickness	Table III	3.2	Gage
104	Length of cap	Table III	3.2	Gage
105	Weight	Table III	3.4	4.5.6
106	Composition	4.4.3.5	3.2	4.5.3
107	Density	4.4.3.7	3.4	4.5.5
108	Diameter of hole	Note 1	3.2	Gage
109	True position of hole	Note 1	3.2	Gage
110	Diameter of rear flat portion	Note 1	3.2	Gage
111	True position of flat portion	Note 1	3.2	Gage
112	Presence of ribs on inside	Note 1	3.2	Gage
113	Nitrocellulose stability	4.4.3.4	3.3	4.5.2
Minor				
201	Improper taper angle	Note 1	3.2	Gage
202	Improper chamfer of hole	Note 1	3.2	Visual
203	Improper chamfer of lip	Note 1	3.2	Visual
204	Improper inside radius	Note 1	3.2	Gage
205	Evidence of poor workmanship	Table III	3.10	Visual

NOTES: 1/ Tool Controlled Dimensions - Whenever a new tool is introduced, 25 samples shall be checked for conformance with the dimension requirements for qualification of tool prior to regular production.

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4.4.3 Testing.

4.4.3.1 50 - lb static test. Ten (10) molded bodies and ten (10) igniter containers shall be randomly selected from the first article sample and from each of the first three production lots and tested in accordance with 4.5.1. If any sample fails to pass the test, the lot shall be rejected. When three consecutive lots have been accepted, sample may be reduced to five (5) molded bodies and five (5) igniter containers from each lot. If a failure occurs, sample shall revert to ten (10) molded bodies and ten (10) igniter containers per lot until three consecutive lots have been accepted. Any failure shall be cause for lot rejection.

4.4.3.2 Assembly test. Each rear cap and igniter container shall be tested in accordance with 4.5.7. If the assembly fails the test, the rear cap and igniter container shall be rejected and removed from the lot (see 6.10).

4.4.3.3 Destructive pull tests. Five (5) rear caps and five (5) igniter containers shall be randomly selected from the lot and tested in accordance with 4.5.8. Any failure shall be cause for lot rejection.

4.4.3.4 Stability. To determine the stability of the molded body, igniter container and rear cap, random samples of three (3) moldings shall be taken from each slurry batch making up a molded body, rear cap or igniter container lot. When three (3) consecutive lots of molded bodies, rear caps, or igniter containers have been accepted, sampling may be reduced to one (1) molding from each batch. In the event of a failure, sample size shall revert back to three (3) per batch. From each such sample, the stability shall be determined in accordance with 4.5.2. Any failure shall be cause for batch rejection.

4.4.3.5 Composition. To determine the composition of the molded body, igniter container and rear cap, one molding shall be withheld and marked immediately after pressing the raw felt from each slurry batch mixture or one molding from the beginning, middle and end of each batch production (when the slurry batch serves for more than one day's manufacture) of the lot. From each such sample, the composition shall be determined in accordance with 4.5.3. Failure of any specimen is cause for rejection of all parts manufactured from that batch since the last previously acceptable analysis and until an acceptable analysis is performed on a subsequently manufactured sample from that batch.

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4.4.3.6 Tensile strength. Five (5) molded bodies shall be randomly selected from the first article and each production lot to provide the required test specimens for tensile tests in accordance with 4.5.4. Failure of any specimen shall be cause for lot rejection.

4.4.3.7 Density. Five (5) molded bodies, five (5) rear caps and five (5) igniter containers shall be randomly selected from the first article sample and each production lot to provide the test specimens for density tests in accordance with 4.5.5. Testing is for informational purpose only.

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 50 - lb static test. The body shall be loaded with a 50 lb inert material and assembled together with the igniter container with the lock-in device. The inert loaded assembly shall be lifted above the ground with the igniter container facing downward and clamped on a suitable test fixture for 5 minutes. Observation shall be made for separation of the igniter container from the body. Any separation occurrence in the testings would be cause for lot rejection.

4.5.2 Stability. The sample quantity of combustible case components shall be subjected to the stability heat test at 134.5 degrees C in accordance with MIL-STD-286, Method 404.1.2.

4.5.3 Composition.

4.5.3.1 Preparation of specimen (see 4.4.3.5.). Each specimen shall be divided into 1/8-inch (approximate) cubes or equivalent. The specimens shall then be thoroughly mixed to form a composite. From the composite, approximately 1.0 g shall be selected as the analysis sample and weighed to the nearest 0.2 milligram (mg), (weight A) in an extraction thimble. The same thimble should be carried through the extractions in 4.5.3.3.1 and 4.5.3.3.3.

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4.5.3.2 Total volatiles. The samples from 4.5.3.1 shall be dried for 3 hours at 100 ± 2 degrees C, cooled in a desiccator (1/2 hr. min.), or placed in a microwave moisture analyzer, dried to constant weight and weighed to the nearest 0.2 mg (weight W). Calculate the percent total volatiles as follows:

$$\% \text{ Total Volatiles} = \frac{A - W \times 100}{A}$$

4.5.3.3 Analytical procedure.

4.5.3.3.1 Extraction. Using a tared, extra-coarse, sintered glass thimble, extract each sample from 4.5.3.2, with methylene chloride, for a minimum of 16 hours in a Roweg extractor (MIL-STD-286, Method 104.1). Evaporate the extract to dryness, using a current of clean, dry air. This extract is to be used for diphenylamine determination in 4.5.3.3.2. Dry the residues (non-extractable material) in the extraction thimbles by aspirating the thimble and contents until the odor of methylene chloride is not discernable. Dry the thimble plus residue for 1/2 hour minimum at 100 ± 20 degrees C. Cool in a desiccator (1/2 hour min). Determine the dry weight of this residue (weight C) to the nearest 0.2 mg as the difference between this weight and the weight of the empty thimble. This residue is to be used for the nitrocellulose determination in 4.5.3.3.3.1.

4.5.3.3.2 Diphenylamine (DPA).

4.5.3.3.2.1 Method 1. To the dried extract from 4.5.3.3.1, add 50 milliliters (ml) of glacial acetic acid. If necessary, warm slightly to dissolve the extract, then cool to room temperature. Complete the determination of diphenylamine in the solution according to Method 201.1 (starting with paragraph 5.5) of MIL-STD-286, except that 25 ml of methylene chloride is to be added after addition of the potassium iodine solution. When the end point appears to have been reached, (absence of blue color) the flask is stoppered and shaken. If no color appears, the end point has been reached; otherwise, continue titration until the blue color is discharged and shaking produces no further color. Calculate the percent DPA as shown in MIL-STD-286, Method 201.4, paragraph 5.11.

4.5.3.3.2.2 Tentative method 2. (Informational test for the first 3 production lots) see 6.8. Prepare 1.2 to 1.5 grams of finely cut sample. Dry in a microwave oven. Transfer the dried sample to a weighing bottle. Cool the weighing bottle in a desiccator for 5 minutes and weigh. Transfer the sample to a

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square beaker. Reweigh the empty weighing bottle. Sample weight = total weight minus weight of empty weighing bottle. Add 50 ml of 90% methanol to the square beaker, grind the mixture by the use of the polytron homogenizer for 3 minutes or until the sample is completely defiberized. Transfer the mixture to a millipore filtering apparatus with 0.45 micron cellulose nitrate membrane. Use a squeeze bottle with 90% methanol to wash the square beaker. Add the washing solution on top of the residue. Add 50 ml of fresh 90% methanol to the square beaker. Grind this solution in the polytron for 1 minute. Use spatula to transfer the residue from the filtering apparatus to the square beaker containing the methanol. Place the beaker in an ultrasonic cleaner which has warm water (minimum 45 degrees C) for 10 minutes. Transfer the mixture to the filtering apparatus without changing the filter. Use a squeeze bottle with 90% methanol to wash the square beaker and add the washing solution to the residue in the filtering apparatus. Repeat steps above two times except the time in the ultrasonic cleaner is reduced to 5 minutes. Transfer the filtrate into a 500 ml volumetric flask. Make up to the mark with 90% methanol. Shake the flask well enough to make sure the solution is uniform. Transfer exactly 15 ml of this solution into a 50 ml volumetric flask. Make up to mark with 90% methanol. Shake well. Measure the UV absorption of this solution at 285 nm. Calculate the DPA content according to the following formula:

$$\text{DPA } \% = \frac{(20A - 0.2) \times 2.5}{32.001 \times W}$$

Where A = absorbance

W = sample weight in gms.

4.5.3.3.3 Nitrocellulose - Schulze - Tiemann Method for Nitrocellulose (NC) determinations.

Assemble the 250 ml Kjeldahl flask, gas evolution apparatus eudiometer/condenser tube, NaOH bath, lead band, heating mantle, transformer and protective shield as shown in Figure I.

Calibrate the apparatus with dry standard A.C.S. grade KN03. Weigh 0.5 - 0.7 g standard sample to 0.1 mg and analyze using the following procedure. The factor is determined using the following equation:

$$F = \frac{616.8\ W\ (273.15 + T)}{V\ (B - C - P - S)}$$

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F = Eudiometer Factor
W = Mass of KNO3 (g)
T = Temperature of Condenser Water (Degrees C)
V = Volume of Gas in Eudiometer Tube (ml)
B = Barometric Pressure (mm Hg), Table IV
C = Temperature Correction (mm Hg), Table IV
P = Vapor Pressure of Water over 33% w/w NaOH at T (mm Hg) Table V
S = Local Gravity & Sea Level Values corrected to standard values

Prepare samples as follows: Cut a specimen into 1/4 inch or smaller squares. Dry to constant weight using microwave moisture analyzer.

Displace the nitric oxide (NO) from the previous run by aspirating the gases through the drain or venting through a hood. Open the stopcock at the top of the eudiometer tube and carefully raise the leveling bulb until all the gas has been expelled. Rest the bulb in its holder and stopper it. Loosen the eudiometer tube from the rubber stopper in the bottom of the bath and drain the NaOH in the tube into the tank. Fill the leveling bulb to 90% of its capacity and place the empty eudiometer tube back onto the rubber stopper and secure it. Fill the eudiometer to the 0.0 ml mark with the leveling bulb, removing all air bubbles, close the stopcock, secure the bulb and stopper it. Replace the 33% w/w NaOH as needed to maintain a constant level in the tank.

Quantitatively transfer 0.8 - 1.0 g (weighed to $\pm$ 0.1 mg) of the dry sample into a 250 ml Kjeldahl flask, add 50 ml deionized water and a few glass beads to prevent bumping. Attach the addition funnel/check value assembly to the freshly greased joint of the Kjeldahl flask. Attach spring clamps or rubber bands to the ears of the flask and addition funnel. Adjust the gas evolution assembly so that the exit tube is gently resting on the bottom of the NaOH bath. Place the lead band round the exit tube opening.

Open the addition funnel stopcock and turn on the heating mantle. Following the purging of the Kjeldahl flask, close the addition funnel stopcock and purge the rest of the tubing for 5-10 minutes. The system is thoroughly purged when no more bubbles escape into the bath.

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Remove the heating mantle from the Kjeldahl flask and observe the drawing of the NaOH solution into the exit tubing and the closing of the check valve. Remove the eudiometer tube from the rubber stopper, place it over the open end of the exit tube. Rest the bottom of the eudiometer tube on the lead band and secure the eudiometer. Refill the leveling bulb to 90% of its volume, stopper the bulb and place it back into its ring.

Place 25 ml of FeCl2 solution into the addition funnel. Open stopcock slowly and allow most of the FeCl2 to flow into the flask. Retain a few drops in the stopcock to preclude the flow of air into the system. If air is allowed into the system, the analysis is invalid and must be repeated. Place 25 ml of concentrated HC1 to the addition funnel and add it to the Kjeldahl flask. Add 10 ml deionized water in a similar fashion. Following the last addition, fill the addition funnel with 30 ml deionized water and tighten the stopcock nut. Place the heating mantle around Kjeldahl flask and heat, for 20-30 minutes or until bubbling has stopped. Remove the heating mantle and check to see that the valve closes. Loosen the stopcock nut on the addition funnel, allow the water to drop into the Kjeldahl, then increase the flow to effectively rinse the walls of the flask, close stopcock before air enters and tighten the nut. Attach the heating mantle, tap the check valve lightly until it opens and continue for 5-10 more minutes.

Following the completion of the reaction, remove the heating mantle and observe the closing of the check valve. Carefully move the eudiometer tube from its present position back to the rubber stopper and secure it. Slowly release the vacuum inside the Kjeldahl flask by allowing water and air to enter the flask through the stopcock.

Tap the check valve to help it open. Allow the gas to equilibrate thermally with the water in the jacket around the eudiometer tube. Wait at least 20 minutes. Measure the volume of gas collected in the eudiometer tube by uncorking the leveling bulb and adjusting it so that the top of the fluid in the bulb is at exactly the same height as the top of the fluid in the eudiometer tube. Read the volume to the nearest 0.1 ml. Use the following equation to calculate the % NC in the sample:

$$NC\ \% = \frac{2.246\ V\ (B - C - P - S)\ F}{NW\ (273.15 + T)}$$

V = Volume of Gas in Eudiometer Tube (ml)
B = Barometric Pressure (mm Hg)
C = Temperature Correction (mm Hg) Table IV

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P = Vapor Pressure of Water over 33% (w/w) NaOH at T (mm Hg) Table IV
F = Eudiometer Factor
N = Percent Nitrogen in NC (%)
W = Mass of Sample (g)
T = Temperature of Condenser Water (Degrees C)
S = Local Gravity & Sea Level values corrected to standard values

PREPARATION OF FeCl2:

Prepare the solution by adding 1000 g of FeCl2.4H20, 1200 ml deionized water and 70 ml of concentrated HCl to a 5 liter round bottom flask. Attach the reflux condenser and reflux for 20 minutes. Cool to room temperature. Slowly add 20 grams of iron filings or powder to the cool solution. When the gas evolution has stopped, filter quickly without aerating the solution to remove the fines. Use A.C.S. grade chemicals throughout the synthesis.

Prepare the bath fluids by dissolving reagent grade NaOH in deionized water until the concentration is 33% weight to weight.

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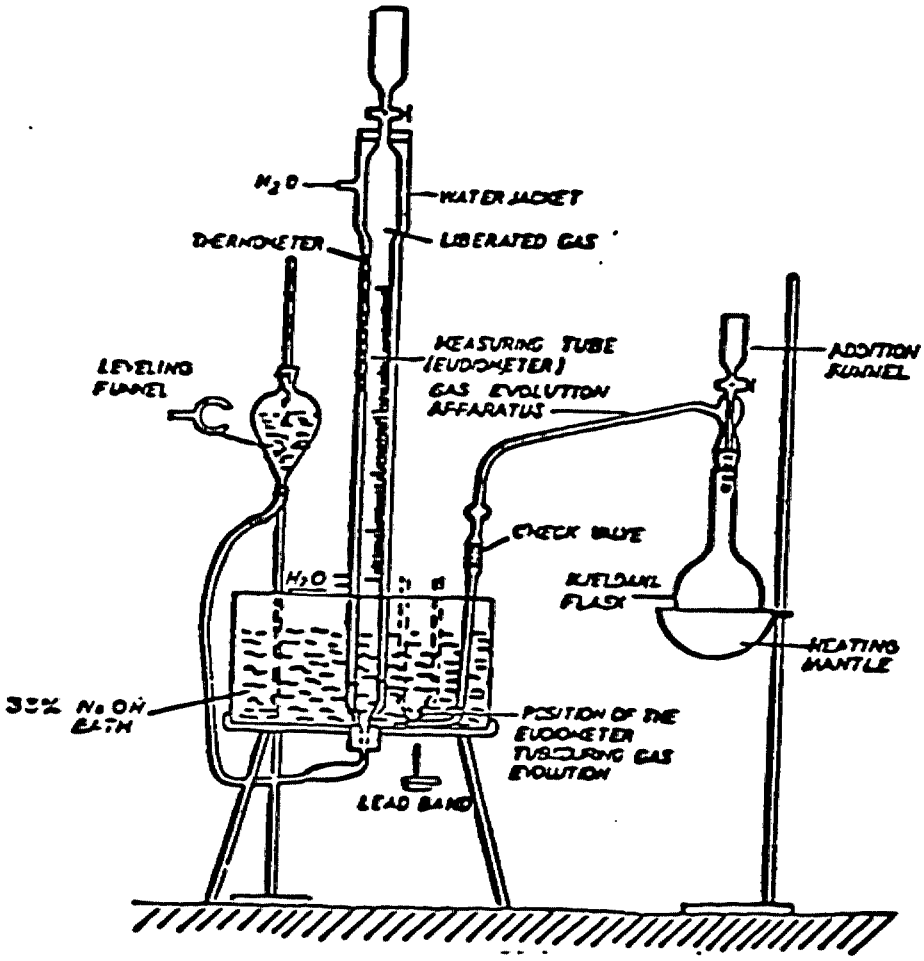


FIGURE 1.

SCHULTZ - TIEMANN - APPARATUS

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TABLE IV - BAROMETRIC PRESSURE IN mm Hg

Temp (See Note 1)	740	760	780
14	1.69	1.73	1.78
15	1.81	1.86	1.91
16	1.93	1.98	2.03
17	2.05	2.10	2.16
18	2.17	2.23	2.29
19	2.29	2.35	2.41
20	2.41	2.47	2.54
21	2.53	2.60	2.67
22	2.65	2.72	2.79
23	2.77	2.84	2.92
24	2.89	2.97	3.05
25	3.01	3.09	3.17
26	3.13	3.21	3.30
27	3.25	3.34	3.42
28	3.37	3.46	3.55

Note 1: Temperature (Degrees C) on Barometer Thermometer.

Vapor Pressure of Water Over 33% (w/w) NaOH - Variable F

Temperature of Condenser Water

(^{o}C)	18	19	20	21	22	23	24	25
P =	11.6	12.4	13.2	14.0	14.9	15.8	16.8	17.8

Barometric pressure temperature correction, Variable C

TABLE V

Pressure correction value due to the vapor pressure of sodium hydroxide solution in the eudiometer tube.

TEMPERATURE (^{o}C)	VAPOR PRESSURE OF NaOH SOLUTION (mm Hg)
18	11.6
19	12.4
20	13.2
21	14.0
22	14.9
23	15.8
24	16.8
25	17.8

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Calculation of P_a (actual pressure) given a temperature T_a (actual temperature):

$$(T_a - T_1) \times (P_2 - P_1) + P_1 = P_a$$

Where: T_a = actual condenser water temperature

T_1 = next lowest tabulated temp from T_a

P_2 = pressure at next highest tabulated temp from T_a

P_1 = pressure at T_1

4.5.3.3.3.1 Nitrocellulose (Alternate method). To provide good heat conduction to the solvent, place a sufficient quantity of glycerin on the flask bottom and the surface of the heat source to produce a continuous film. Add 10 ml of ethyl alcohol to the thimble and residue from 4.5.3.3.1 and extract with acetone using the method in 4.5.3.3.1. After extraction dry the residue remaining at 100 ± 2 degrees C for one hour. Cool in a desiccator (1/2 hour min) weigh to the nearest 0.2 mg (Weight D). Evaporate the acetone extract to dryness using the inert atmosphere throughout this test. Reflux the solution for five (5) minutes. Add to the flask, via the condenser, 25 ml of 0.7N ferrous ammonium sulfate (prepared per MIL-STD-286, Method 604.1) and 10 ml concentrated hydrochloric acid. Reflux for thirty (30) minutes more to obtain a yellow liquid in which the specimen is completely disintegrated. With the inert gas still running, cool in a water bath to room temperature. Add 5 ml of 20% ammoniumthiocyanate solution (MIL-STD-286, Method 601.1). Perform a blank titration on the reagents. Calculate the percent NC as follows:

$$\% \text{ NC} = \frac{46.69N \quad (V-B) \quad X \ 100}{E \quad W}$$

V = Volume titanous chloride used for sample, ml
B = Volume titanous chloride used for blank, ml
W = Weight of dried sample, gm
N = Normality of titanous chloride
E = Percent nitrogen in the nitrocellulose

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4.5.3.3.4 Talc content (ash test). The specimen of approximately 1 - 1.5 grams shall be taken from the same case body for the above testing and weighed to within 0.2 mg in a tared porcelain crucible. Cover the crucible. The specimen shall then be ignited at 500 degrees C in a muffle furnace. The crucible shall be cooled to room temperature and then weighed. Heat until all evidence of carbon is gone (no black residue), for one hour. The increase in weight of the crucible shall be calculated on a total volatile - free basis to the percent of talc content.

4.5.3.3.5 Kraft, resin and additives. Calculate the percent kraft, resin and additives as follows:

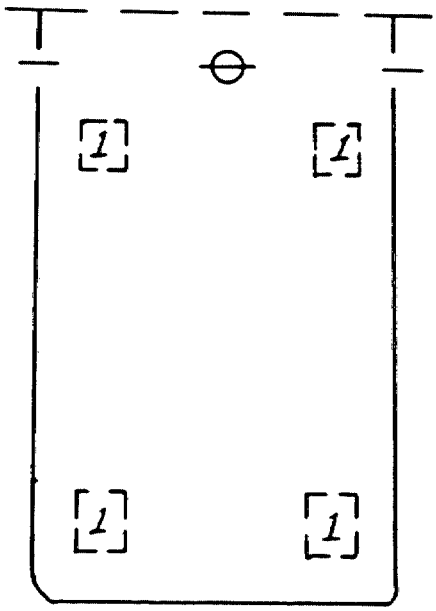
% Kraft + % resin + % additives = 100 - (% NC + % DPA + % Talc)

4.5.3.3.6 Report the total methylene chloride extractables $\frac{(W - C) \times 100}{W}$ and the total acetone extractables $\frac{(C - D) \times 100}{W}$.

4.5.4 Tensile strength. Two specimens shall be prepared from each molded body. The specimens shall be taken from the center of the wall of the body. Test specimens shall be taken 180 degrees apart. The specimens shall have a one-inch gage length and width, but wall thickness shall fall within the tolerance specified on the applicable drawing. The test specimens shall be preconditioned for a minimum of 4 hours at 23 plus or minus (±) 2 degrees centigrade (C) at a relative humidity of 50 ± 5 percent (%). The tensile breaking strength of the specimens shall be determined in accordance with ASTM-D-638, using a cross head speed of one (1) inch per minute.

4.5.5 Density. Six (6) specimens shall be taken from each molded body in accordance with Figure 2. Three (3) specimens shall be taken from each rear cap and igniter container sample in accordance with Figure 3 and Figure 4 respectively. All the specimens shall be dried for 3 hours at 105 degrees C or for 12 hours at 70 degrees C or dried to constant weight in a microwave moisture analyzer. The specimen thickness shall be measured, the volume computed and the sample weighed on a calibrated scale having an accuracy of plus/minus 0.02g. The average density for each part shall be calculated and shall be in accordance with the applicable requirements.

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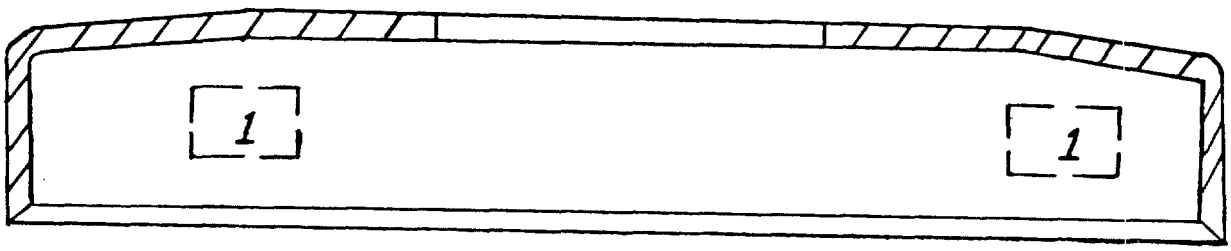


1. Three samples, 1 in. X 3 in. displaced by 120 degrees for testing the density.

(Each from top and bottom)

FIGURE 2. Body

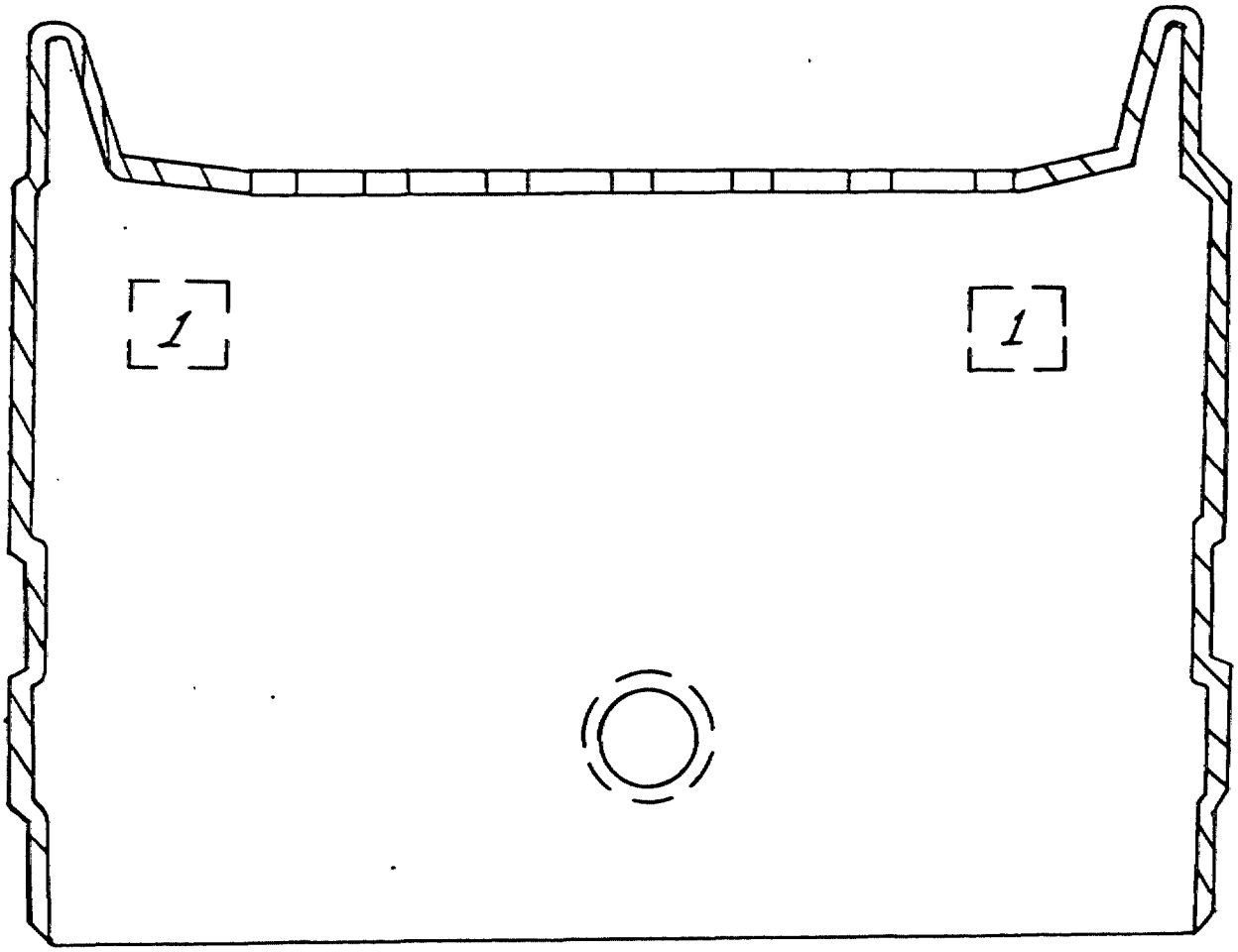
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1. Three samples, 1/2 in. by 1/2 in. displaced by 120 degrees for density test.

FIGURE 3. Rear Cap

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1. Three samples, 1 in. X 1 in. displaced by 120 degrees for density test.

FIGURE 4. Igniter Container

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4.5.6 Weight. All samples shall be weighed individually using any Government approved and suitably calibrated scale capable of reading to the nearest 0.1 g. Test samples shall be preconditioned for a minimum of 4 hours at 23 ± 2 degrees C and 50 ± 5 % R.H. prior to weighing.

4.5.7 Assembly test. The rear cap shall be assembled to igniter container. The assembly shall be lifted above the surface by using fingers through the hole of the rear cap without touching the igniter container. Separation of the igniter container from the rear cap shall be cause for rejection of the assembly. The rejected assembly shall be removed from the lot for proper disposition (See 6.10).

4.5.8 Destructive pull test. This test shall be performed after the adhesive bond between the igniter container and rear cap has had sufficient time to set. The assembly shall be held in a suitable test fixture clamping on the igniter container. The test apparatus shall apply a longitudinal load distributed over 360 degrees around the lip of the rear cap. Observation shall be for the force required for separation of the rear cap from the igniter container. If any sample fails to meet the specified requirement, the lot shall be rejected.

4.5.9 Proving ground firing. The combustible case components shall be assembled as 155mm M203A1 propelling charge with the established service charge load in accordance with MIL-C-70462. The tests shall be conducted at a Government proving ground in accordance with the applicable acceptable test procedure using M549, RA or PXR-6325 projectiles inert loaded within 0.1 pound of an as fired weight of 96.0 pounds using primer, M82 and fired in cannon M199 with 75 percent (minimum) remaining life. Firing shall be conducted with a spindle to charge base stand-off of 1 inch (plus or minus 1/4 inch). Prior to firing, all charges shall be temperature conditioned in a large conditioning box with adequate air circulation at the required temperature for a minimum of 24 hours and assurance will be made that, when fired, the charges are at the conditioning temperature. The following tolerances shall apply to all -65 degrees F, from -65 degrees F to -60 degrees F; 145 degrees F from 140 degrees F to 145 degrees F; 70 degrees F from 68 degrees F to 72 degrees F. The test samples shall be fired at the temperatures indicated in Table VI and any inert or burning residue shall be cause for lot rejection. Camera coverage of the weapon chamber and tube shall be provided for all test firings.

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TABLE VI. Proving ground firing

Temperature	First Article	First Lot or Each New Interfix
- 65 °F	10	10
+ 70 °F	10	10
+145 °F	10	10

4.5.9.1 Refiring. If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the results shall be declared invalid, and a new test shall be performed.

5. PACKAGING

5.1 Packing.

5.1.1 Packing for igniter container and rear cap.

5.1.1.1 Level A. The Igniter Container and Rear Cap shall be packed in accordance with drawing 9390475-1.

5.1.1.2 Level B. The Igniter Container and Rear Cap shall be packed in accordance with drawing 9390475-2.

5.1.1.3 Level C. The Igniter Container and Rear Cap shall be packed in accordance with drawing 9390469.

5.1.2 Packing for body, molded.

5.1.2.1 Level A. The Molded Body shall be packed in accordance with drawing 9390476-1.

5.1.2.2 Level B. The Molded Body shall be packed in accordance with drawing 9390476-2.

5.1.2.3 Level C. The Molded Body shall be packed in accordance with drawing 9390470.

5.2 Marking.

5.2.1 Marking for igniter container and rear cap.

5.2.1.1 Level A and B. Mark in accordance with drawing 9390475.

5.2.1.2 Level C. Mark in accordance with drawing 9390469.

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5.2.2 Marking for body, molded.

5.2.2.1 Level A and B. Mark in accordance with drawing 9390476.

5.2.2.2 Level C. Mark in accordance with drawing 9390470.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

6.1 Intended use. The items covered by this specification are intended for use in M203A1 155MM Propelling Charge.

6.2 Acquisition requirements. Acquistion documents must specify the following (see MIL-A-48078).

a. Title, number, and date of the specification.
b. Issues of DoDISS to be cited in the solicitation and if required, the specific issue of the individual documents referenced (see 2.1)
c. Provisions for submission of first article sample.

6.3 Submission of designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required to Commander, ARDEC, ATTN: AMSMC-QAR-I(D), Picatinny Arsenal, NJ 07806-5000.

6.4 Submission of ammunition data cards. In addition to the normal distribution of records, when the cases are procured by the Army, one copy of all ammunition data cards shall be forwarded to each of the following:

Commander
ARDEC
ATTN: AMSMC-QAR-R(D)
SMCAR-AEE-BP
Picatinny Arsenal, NJ 07806-5000

6.5 Excessive residue. Residue is defined as excessive if it is significant enough to prevent chambering or hinder functioning of a subsequent round, or exceed 1 inch by 2 inches in size, as determined by the proving ground.

6.6 Proving ground.

6.6.1 Cleaning. Prior to each firing, the gun breech shall be thoroughly cleaned and dried.

6.6.2 Handling and storage. Special care shall be taken in the handling and storage of sample cases to prevent contamination by grease, grime, dirt, or other foreign material.

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6.7 First article test. The first article test is a one time qualification test for a new contractor to produce acceptable combustible cases. Also, the charge weight adjustment of using the new contractor's combustuble case against the combustible cases used in the calibration components shall be determined in the first article test. Once a contractor has demonstrated the ability to produce acceptable combustible cases and the charge weight adjustment of using the new contractor's combustible cases has been determined, the first article test may be waived by the contracting officer. The following test program shall be used for the first article test:

TABLE VII. First article test

Test Phase	Temp o (F)	No. of Control Rd	No. of Test Rd	Test Procedure
Sequential Environmental	145	15	60	TOP-4-2-504
	-60	15	60	TOP-4-2-504
Charge weight adjustment test	See Table VIII.			

TABLE VIII. Charge weight adjustment test

Test Phase	Gun Tube No.	Temp o (F)	Sample Size Cal	Test (See Note 1)
Charge Establishment	1	70	3	9 (See Note 2)
	1	70	3	9 (See Note 3)
Uniformity	1	70	10	10 (See Note 2)
	1	70	10	10 (See Note 3)
	1	145		10 (See Note 2)
	1	145		10 (See Note 3)
	1	-65		10 (See Note 2)
	1	-65		10 (See Note 3)
	2	70	10	10 (See Note 3)
	3	70	10	10 (See Note 3)

Note 1: One new lot of M31A1E1 propellant shall be used for the charge weight adjustment test. Thirty boxes of propellant shall be randomly selected from the lot and shipped to the proving ground for charge weight adjustment test. Three gun tubes shall be used for the test. The test shall provide the two recommended charge weights. The first recommended charge weight shall be

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based on the established calibration combustible cases used for the test charges. The second recommended charge weight shall be based on the average of the three recommended charge weights using the new contractor's combustible cases. The difference between the first and second recommended charge weight will be used for the charge weight adjustment for the new contractor's combustile cases. The test shall be done in accordance with MIL-P-70456.

Note 2: Calibration combustible cases shall be used for the test charges.

Note 3: New contractor's combustible cases shall be used for the test charges.

6.8 Informational DPA content determination. The tentative Method 2 for DPA content determinations is for informational purposes and cannot be used for acceptance test purposes. The informational test will be done for the first 3 production lots in addition to the normal acceptance test Method 1. All the above batch composition test results shall be submitted to Commander, ARDEC, ATNN: AMSMC-QAR-R(D), Picatinny Arsenal, NJ 07801-5000 for statistical evaluation of equivalency of both methods. If the tentative Method 2 is found to be equivalent to Method 1, an ECP will be submitted for changing the tentative method to alternate method in this specification.

6.9 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions should be submitted by the contractor for evaluation by the technical activity responsible for preparation of this specification.

6.10 Assembly test. The rear cap and igniter container which fail the assembly test may be used again and can be assembled to different rear cap or igniter container with the same interfix number. The failed rear cap and igniter container will not be considered acceptable until they pass the assembly test.

6.11 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDEC) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, ARRADCOM or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

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6.12 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.13 Subject term (key word) listing.

Molded Body
Rear Cap
Igniter Container
Nitrocellulose

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1320-AC31)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-70461D (AR)	28 May 1991

3. DOCUMENT TITLE
COMBUSTIBLE CARTRIDGE CASE ASSEMBLY PARTS FOR CHARGE, PROPELLING, 155MM

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6675	(2) AUTOVON 880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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